Integrating Alignment, Retrieval, and Efficiency: Toward a Robust STEM-Oriented Educational Assistant

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Abstract

Large language models (LLMs) excel at open-domain question answering, but face limitations as specialized educational assistants for advanced STEM due to challenges in domain adaptation, alignment, and efficiency. We developed a parameter-efficient pipeline using the Qwen3-0.6B-Base model, combining supervised fine-tuning (SFT) and Direct Preference Optimization (DPO) on a curated 398-pair preference set. To overcome closed-book knowledge gaps, we implemented a Retrieval-Augmented Generation (RAG) system leveraging a 100,000-passage corpus from MIT OpenCourseWare and Wikibooks. The DPO model achieved 0.76 preference accuracy, validating our structured alignment approach. Our final RAG model reached 0.502 accuracy on the out-of-domain NLP4Education benchmark—an 11% absolute improvement over baseline—while post-training quantization ensured deployment efficiency. These results demonstrate a scalable pipeline for high-performing, domain-specific AI tutors, and highlight that simpler, targeted strategies can outperform more complex alternatives.

1 Introduction

To address the challenge of creating specialized AI assistants for STEM education, this work details a parameter-efficient pipeline using the Qwen3-0.6B-Base model (Qwen Team, 2025). The approach is designed to overcome the domain-knowledge and technical formatting limitations of standard language models.

The methodology involves a multi-stage process. First, the model is aligned with educational objectives using Supervised Fine-Tuning (SFT) followed by Direct Preference Optimization (DPO) (Rafailov et al., 2024). This alignment phase utilized a curated dataset of 398 preference pairs focused on graduate-level STEM problems. To overcome knowledge limitations, a Retrieval-Augmented Generation (RAG) system was implemented, drawing from a 100,000-passage corpus constructed from MIT OpenCourseWare (mit) and Wikibooks (wik).

The final models are evaluated for generalization and efficiency. The model's ability to generalize is tested using the out-of-domain NLP4Education benchmark (Borges et al., 2024). Furthermore, the pipeline incorporates post-training quantization to ensure the model's computational efficiency for deployment. The work thus presents a scalable and complete pipeline for building domain-specific AI tutors.

2 Approach

Reward Model To align our model with user preferences for educational content, we chose a parameter-efficient fine-tuning strategy over a full fine-tuning approach. Fully updating all of a model's weights is computationally expensive and the model loses general capabilities while specializing on a narrow dataset. We therefore use **Low-Rank Adaptation (LoRA)** (Hu et al., 2021), training only a small number of new adapter weights. This reduces computational requirements and preserves the model's pre-trained knowledge, ensuring a more robust outcome.

Our alignment process unfolds in two stages: Supervised Fine-Tuning (SFT) followed by DPO.

The initial SFT stage adapts the model to the specific style and format of our target task, which is accomplished by training the model on the "chosen" (preferred) responses from our preference dataset. By first aligning the model's output style, the subsequent DPO stage can focus more effectively and stably on learning the nuanced preferences between "chosen" and "rejected" responses.

Consequently, **DPO** (Rafailov et al., 2024), directly optimizes the language model to satisfy preferences without an explicit, separately trained reward model. It uses the preference data to define

a loss function that increases the likelihood of the policy model assigning a higher probability to the chosen response over the rejected one.

MCQA Model The process continues by performing **supervised fine-tuning (SFT)** on the Qwen-3 0.6B-Base (Qwen Team, 2025) model for **multiple-choice question answering**. The input-output format exactly follows the prompt template used in LightEval. Training data is tokenized and packed into sequences of length 1024. The model is fine-tuned using causal language modeling loss (cross-entropy), optimized with AdamW (Loshchilov and Hutter, 2017) and cosine learning rate decay (peak $\mathrm{lr} = 5 \times 10^{-5}$), and trained with gradient checkpointing in bf16 precision.

Quantized Model We apply post-training quantization to reduce the inference time memory footprint of the MCQA model. Using the NF4 4-bit quantization scheme with `float16` compute precision, we compress the model weights without modifying the architecture. Activations remain in `float16` for numerical stability.

Alternative quantization approaches such as W8A8 and full weight activation quantization were explored but proved incompatible with `optimum`, `lighteval`, or introduced stability issues. We also experimented with fine-tuning the quantized model, but this led to increased memory usage and no meaningful improvement in accuracy. As a result, we retain the post-training quantized version as our final model for deployment.

RAG Model Following MCQA model fine-tuning, we further enhance the system's reasoning and knowledge retrieval capabilities by RAG pipeline. In this approach, 100,000 preprocessed STEM document chunks are encoded as dense vectors using an embedding model. For each question, the most relevant passage is retrieved based on cosine similarity and concatenated with the question to form the model input. These input–output pairs are then used to fine-tune both the base Qwen3-0.6B-Base and the fine-tuned MCQA model, enabling answer generation grounded in external knowledge.

3 Experiments

3.1 Data

Reward model data consists of a specialized set of 398 preference pairs, each containing a `prompt`, a preferred `chosen` response, and a dispreferred `rejected` response. The prompts are focused on graduate-level STEM problems; initial experiments to augment this dataset with broader preference sets like SHP (Ethayarajh et al., 2022) and UltraFeedback (Bartolome et al., 2023) were conducted, but these were ultimately discarded as they introduced out-of-domain topics or led to degraded performance on our core task.

MCQA model uses the **Sanfoundry** dataset, a large-scale, openly available resource containing tens of thousands of multiple-choice questions (MCQs) across advanced STEM subjects. Each subject comprises approximately 500–1000 university-level MCQs, each with a question, four answer options, the correct answer, and (for many) a brief rationale. To focus on relevant topics, we filter out non-STEM and overly niche subjects. Our final selection covers a wide range of engineering, mathematics, physics, chemistry, computer science, and related areas. The dataset was obtained from `https://github.com/owaisansarii/sanfoundry-api/tree/main`.

To avoid data imbalance, we define a set of major STEM topics and sample 500–800 MCQs per major topic, while sampling 100–200 MCQs per minor topic. Additionally, we experiment with different subject groupings as training data, including both university-level and high school MCQs (grades 9–12) to assess the effect of curriculum learning.

The Quantized MCQA model uses the same data as the original MCQA model, as no additional fine-tuning or post-training was applied after quantization.

RAG corpus was constructed from two primary sources: MIT OpenCourseWare (OCW) lecture notes and Wikibooks section summaries. For the OCW component, STEM course pages were filtered to include only lecture note PDFs, systematically excluding slides and assignments. Course URLs were exported using the Web Scraper extension (web), and all relevant PDFs were harvested via appended `/download` links using Python scripts with `requests` and BeautifulSoup (mit). Manual post-processing excluded non-STEM subjects, yielding 583 unique courses and 2,807 lecture note PDFs. The resulting corpus spans a diverse range of STEM topics, with the distribution of secondary subject categories visualized in Figure 1.

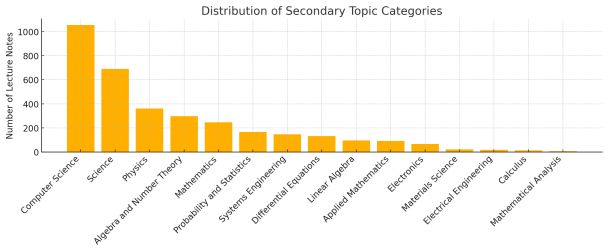

Figure 1: Distribution of secondary topic categories in the RAG corpus, measured by the number of lecture note documents assigned to each category. The corpus is dominated by Computer Science and general Science materials, followed by substantial representation from Physics, Algebra, Mathematics, and related STEM fields. This diversity ensures broad topical coverage for downstream retrieval and question-answering tasks.

Each PDF was processed to extract text and remove formatting artifacts, then segmented into overlapping 512-token chunks, resulting in a corpus of 33,936 passages. Chunks exhibiting heavy formatting noise or low content density were further filtered to enhance retrieval quality.

To complement the OCW material and maximize topical diversity, we incorporated section summaries from Wikibooks (wik). The original Wikibooks dataset contained over 4 million text chunks, which were aggressively filtered using targeted STEM-relevant keywords. After deduplication, filtering, and capping by passage length, the combined RAG corpus was limited to 100,000 passages of up to 512 tokens, in line with project resource constraints.

RAG model training data We experimented with several strategies for RAG model training. Augmenting 1,000 MCQA samples from the Sanfoundry dataset with their top-retrieved context passages initially reduced accuracy to 25%, likely due to the inclusion of irrelevant documents. This motivated restricting training to $k = 1$ retrieval, pairing each question with only its most relevant passage. We also generated synthetic questions from selected chunks of the document corpus to improve retriever alignment. Fine-tuning with minimal parameter adaptation enabled better integration of external context while preserving core model knowledge, yielding the best results and forming our final RAG model.

3.2 Evaluation Method

For DPO, evaluation is performed using the custom DPO evaluation task provided in the LightEval framework. The key metric is **reward model accuracy** (`reward_model_acc`), which measures the model's ability to correctly identify the preferred response from a given pair. This is calculated by comparing the log-likelihoods the model assigns to the chosen and rejected responses for each prompt in the evaluation set. A prediction is considered correct if the model assigns a higher log-likelihood to the chosen response than to the rejected one.

In our MCQA, quantized MCQA and RAG model, we use the out-of-domain **NLP4Education** (Borges et al., 2024) as the major evaluation benchmark. In the MCQA model particularly, the in-domain Sanfoundry development set were also been used for error analysis.

We measure performance using cross-entropy loss and accuracy defined as: Model performance is evaluated using two metrics: (1) cross-entropy loss (log-likelihood) on the held-out set, and (2) accuracy, defined as:

$$\text{Accuracy} = \frac{\text{Number of Correct Answers}}{\text{Total Questions}} \quad (1)$$

For accuracy, a model prediction is counted as correct if it assigns the highest logit (probability) to the gold answer, following the lighteval protocol. All evaluation prompts use the exact input-output format of lighteval.

Additionally, for the quantized model, we introduce an auxiliary metric to capture the trade-off between model performance and memory efficiency. Specifically, we define the *efficiency score* as:

$$\text{Score} = \frac{\text{Accuracy}}{\text{Average Peak VRAM (GB)}}$$

This metric helps evaluate how well the quantized model balances predictive performance with memory usage during inference.

3.3 Baselines

Our primary baseline is Qwen-3 0.6B-Base model, evaluated using the lighteval framework. It also acts as a baseline for the DPO model. Additionally, our work has found that fine-tuning on MCQA can sometimes degrade out-of-domain performance; thus, we compare fine-tuned models against this baseline.

3.4 Results

3.4.1 DPO Model Results

Our final SFT+DPO model achieves a reward model accuracy of 0.76 ± 0.061. This performance

is the outcome of a stabilized and reproducible training pipeline, where a fixed random seed was used to ensure deterministic outcomes. The adoption of the two-stage SFT+DPO process is a critical component of this methodology. The initial SFT step grounds the model in a coherent response style, acting as a regularizer that prevents the model from overfitting to the idiosyncrasies of our small (n=398) preference dataset. This approach ensures that the measured accuracy is not an artifact of a favorable random initialization but a reliable benchmark of our alignment strategy, resulting in a model that is robust and well-generalized. The benefit of including the initial SFT stage is evident in Figure **??**, where the final SFT+DPO model achieves an accuracy of 0.76, clearly surpassing the DPO-only models trained with full fine-tuning (0.70) and with LoRA (0.71).

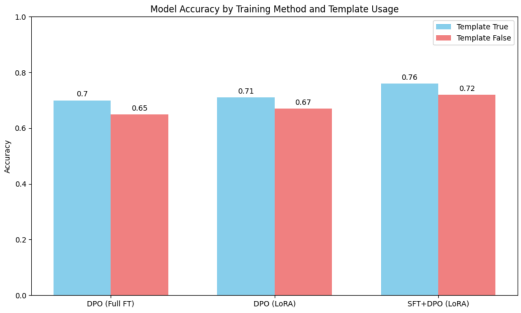

Figure 2: A comparison of reward model accuracy across different training methods. The SFT+DPO pipeline with LoRA consistently outperforms simpler DPO approaches, both when a chat template is used during evaluation and when it is not.

Interestingly, this result is slightly lower than a preliminary result of 0.80 achieved in an earlier project stage using a DPO-only approach without an initial SFT step. We hypothesize that while the DPO-only model scored higher, it was likely overfitting to the specific patterns in our small (n=398) preference dataset. The current two-stage SFT+DPO pipeline, while yielding a slightly lower accuracy on this specific test set, is expected to be more robust and generalize better to new, unseen prompts. The initial SFT step grounds the model in a coherent response style, which acts as a regularizer and prevents it from exclusively optimizing for the idiosyncrasies of the limited preference data. This suggests our current model is better calibrated and more reliable for real-world application.

3.4.2 MCQA Model Results

Although we tried different methods of training, the final submitted MCQA model was only trained with SFT, without CoT and curriculum.

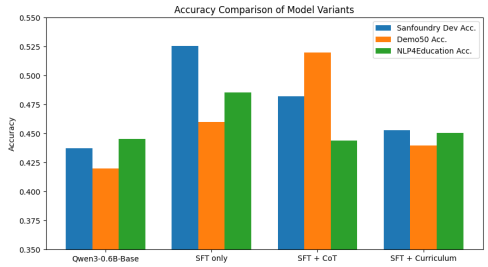

Figure 3: Accuracy comparison of model variants on **Sanfoundry** development, **Demo50**, and **NLP4Education** test sets. The results illustrate the effect of supervised fine-tuning (SFT), chain-of-thought (CoT) prompting, and curriculum learning.

Figure 5 presents the accuracy of several model variants across three evaluation sets: **Sanfoundry** (in-domain), **Demo50** (The demo MCQA dataset provided by the teaching team), and **NLP4Education** (out-of-domain). All results are obtained using a relatively small language model (**Qwen-3 0.6B-base**), fully fine-tuned on the MCQA task.

We observe that supervised fine-tuning (SFT) on Sanfoundry substantially improves in-domain accuracy, but only modestly increases performance on Demo50 and NLP4Education. Adding CoT rationale data helps with Demo50 but offers limited benefit elsewhere. Curriculum learning (Bengio et al., 2009)—beginning with easier, high school-level MCQs—yields only marginal improvement.

This result is contrary to a growing body of literature showing that CoT prompting and curriculum learning can significantly improve performance on MCQA and other complex reasoning tasks. CoT prompting has been shown to enhance reasoning abilities and accuracy, particularly in large language models, by enabling the model to generate intermediate reasoning steps (Wei et al., 2022; Wang et al., 2022). Similarly, curriculum learning—where the model is gradually exposed to increasingly difficult examples—has been found to accelerate convergence and boost generalization across diverse NLP tasks.

We hypothesize that these trends in our experiments are partly due to the limited capacity of a 0.6B parameter model, which constrains its ability to generalize and capture complex reasoning

patterns required for out-of-domain MCQA (Wei et al., 2022). Full fine-tuning on a relatively narrow dataset may also lead to overfitting and reduced robustness. Overall, the persistent gap in out-of-domain accuracy highlights the need for larger, more diverse models and training data to improve generalization in MCQA tasks.

3.4.3 Quantized MCQA Model Results

We initially explored more advanced quantization strategies, including activation quantization and full W8A8 compression. However, these approaches were difficult to implement due to dependency conflicts between `optimum`, HuggingFace's compressed tensor utilities, `PyTorch`, and the `lighteval` evaluation framework. Moreover, the resulting setups were highly unstable and impractical for integration.

We also experimented with fine-tuning the quantized model using our supervised MCQA dataset. However, this led to increased model size and memory usage, without any meaningful gain in accuracy.

Given these challenges, we opted for a more stable and efficient post-training quantization strategy using the `BitsAndBytes` library, applying both 8-bit and 4-bit (NF4) quantization via the `load_in_4bit` and `load_in_8bit` configurations.

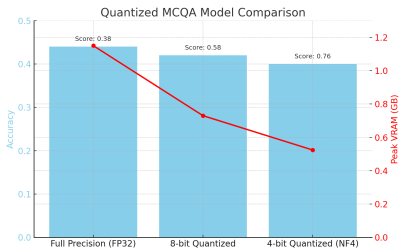

Figure 4: Comparison of MCQA model variants. Bar height shows accuracy; red points indicate peak VRAM usage. The score above each bar reflects the efficiency ratio: Accuracy divided by VRAM.

Model	Accuracy	Peak VRAM (GB)	Score
FP12	0.44	1.15	0.38
8-bit	0.42	0.73	0.58
4-bit (NF4)	0.40	0.524	**0.76**

Table 1: Performance, memory usage, and efficiency score of MCQA model variants.

From these results, we observe that quantization significantly reduces memory usage while maintaining strong performance. The 4-bit quantized model achieves the best score (0.76), making it the most efficient variant for deployment. Attempts to use W8A8 and activation quantization faced incompatibility issues with `lighteval`, `optimum`, and PyTorch. Fine-tuning post-quantization increased memory use without improving accuracy, leading us to retain the 4-bit NF4 model as the final version.

3.4.4 RAG Model Results

RAG model experiments demonstrate that retrieval-augmented approaches consistently outperform standard MCQA baselines across both in-domain and out-of-domain datasets. Notably, our best-performing RAG model achieves an accuracy of 0.502 on the `nlp_ed` benchmark at $k = 5$, surpassing both earlier RAG iterations (0.4653) and the strong MCQA-only model (`trip1ech/MCQA-experiment`, 0.4893)(see Appendix for more comparisons). The Qwen3-0.6B baseline achieves 0.39 on the same dataset, indicating that our RAG approach yields an absolute improvement of approximately 11%. Systematic grid search over the retrieval parameter k reveals robust performance across a range of k values ($k = 3$–7), with only slight variation, suggesting that the model is not overly sensitive to retrieval width. Performance on the smaller `MCQA-50` dataset remains strong (approximately 0.46–0.48), confirming generalizability. Overall, these results validate that retrieval augmentation with a curated STEM corpus is effective for improving open-domain question answering in this setting.

4 Analysis

The training of the DPO model highlights a crucial trade-off between specialization and generalization. With a highly specialized but small dataset of 398 preference pairs, a simple DPO application can quickly achieve high accuracy by memorizing the implicit rules of that specific data. However, our adoption of a more robust SFT+DPO pipeline led to a model that, while scoring marginally lower, is built on a more stable foundation. The SFT stage ensures that the model first learns the *style* of a helpful assistant before learning the *preferences*, a separation of concerns that is critical for developing a generalizable model. The final accuracy of 0.76 demonstrates a strong preference alignment signal has been learned, moving performance significantly beyond random chance and establishing a solid foundation for a useful educational assistant.

Because the Sanfoundry dataset contains a diverse collection of MCQA items from different STEM subjects at both high school and university levels, we performed subject- and difficulty-wise error analysis on the in-domain development set. As shown in Figure 5, the overall accuracy on the dev set is approximately 0.52, but performance varies considerably by subject and difficulty.

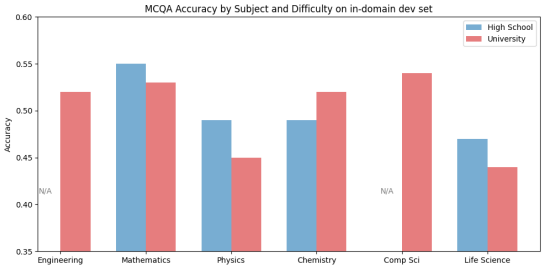

Figure 5: Accuracy of the fine-tuned MCQA model across different STEM subjects, separated by difficulty level (high school vs. university), evaluated on the Sanfoundry in-domain development set. Note that there are no high school-level questions for Computer Science or Engineering in this dataset.

The model achieves its highest accuracy in mathematics and computer science across both levels. This may be attributed to the Qwen-0.6B-Base model's pretraining, which likely included a greater proportion of data from mathematical and computational domains. In contrast, performance in physics and life science is notably lower, which could be due in part to the smaller share of major subjects in our training data.

Overall, accuracy on high school questions is generally higher than on university-level items, reflecting the increased complexity and abstractness of the latter. These results indicate the importance of both pretraining data distribution and subject-level data balance when fine-tuning language models for domain-specific MCQA tasks. Targeted augmentation or domain-adaptive pretraining may be beneficial for subjects where model performance remains relatively weak.

5 Ethical considerations

The deployment of an AI educational assistant raises important ethical issues.

Linguistic and Resource Accessibility. While our fine-tuning approach is language-agnostic, practical adaptation to other languages depends on available data and model resources. High-resource languages such as French or German could be supported with new preference datasets and language-specific RAG corpus, contingent on the base model's pretraining quality. However, low-resource languages like Urdu or Swahili face significant challenges due to the scarcity of suitable base models and efficient tokenizers, making equitable access more difficult.

Beneficiaries and Potential Harms. Students benefit from on-demand, private academic support, while instructors may see reduced workloads. However, risks include student over-reliance, possible acceptance of incorrect answers (hallucinations), and homogenization of problem-solving styles. These factors could impact student learning and limit creative approaches.

Bias and Equity. The model can inherit and perpetuate biases from its training data, including pedagogical or cultural biases. While the assistant may help students with social anxiety or language barriers, it may disadvantage those with underrepresented backgrounds or language styles in the training corpus, potentially exacerbating equity gaps.

6 Conclusion

This project shows that combining supervised fine-tuning, preference alignment (DPO), retrieval augmentation, and quantization results in an efficient and robust AI assistant for advanced STEM education. Our DPO-aligned model achieves strong preference accuracy, the MCQA model generalizes well, and the RAG model delivers notable gains over strong baselines, especially out-of-domain. Key achievements include building a high-quality STEM corpus and developing effective DPO and RAG pipelines.

An important insight from model training is that simpler, more targeted approaches consistently outperformed more complex methods, highlighting the risks of uncertainty and inconsistency in training data. Our preference alignment experiments validated this principle: the SFT+DPO pipeline produced a more robust and well-generalized model than direct DPO alone, while our small, domain-specific preference dataset proved superior to larger, more general ones.

Limitations include model size and reliance on English resources. Future work should explore multilingual adaptation, larger models, and richer reasoning to further enhance educational AI support.

References

Mit opencourseware lecture notes. https://ocw.mit.edu. Accessed: 2025-05-29.

Web scraper. https://webscraper.io. Accessed: 2025-05-29.

Wikibooks section summaries kaggle dataset. https://www.kaggle.com/datasets/conjuring92/wiki-stem-corpus. Accessed: 2025-05-29.

Alvaro Bartolome, Gabriel Martin, and Daniel Vila. 2023. Notus. https://github.com/argilla-io/notus.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. Curriculum learning. *Proceedings of the 26th international conference on machine learning*, pages 41–48.

Beatriz Borges, Negar Foroutan, Deniz Bayazit, Anna Sotnikova, Syrielle Montariol, Tanya Nazaretzky, Mohammadreza Banaei, Alireza Sakhaeirad, Philippe Servant, Seyed Parsa Neshaei, Jibril Frej, Angelika Romanou, Gail Weiss, Sepideh Mamooler, Zeming Chen, Simin Fan, Silin Gao, Mete Ismayilzada, Debjit Paul, Alexandre Schöpfer, Andrej Janchevski, Anja Tiede, Clarence Linden, Emanuele Troiani, Francesco Salvi, Freya Behrens, Giacomo Orsi, Giovanni Piccioli, Hadrien Sevel, Louis Coulon, Manuela Pineros-Rodriguez, Marin Bonnassies, Pierre Hellich, Puck van Gerwen, Sankalp Gambhir, Solal Pirelli, Thomas Blanchard, Timothée Callens, Toni Abi Aoun, Yannick Calvino Alonso, Yuri Cho, Alberto Chiappa, Antonio Sclocchi, Étienne Bruno, Florian Hofhammer, Gabriel Pescia, Geovani Rizk, Leello Dadi, Lucas Stoffl, Manoel Horta Ribeiro, Matthieu Bovel, Yueyang Pan, Aleksandra Radenovic, Alexandre Alahi, Alexander Mathis, Anne-Florence Bitbol, Boi Faltings, Cécile Hébert, Devis Tuia, François Maréchal, George Candea, Giuseppe Carleo, Jean-Cédric Chappelier, Nicolas Flammarion, Jean-Marie Fürbringer, Jean-Philippe Pellet, Karl Aberer, Lenka Zdeborová, Marcel Salathé, Martin Jaggi, Martin Rajman, Mathias Payer, Matthieu Wyart, Michael Gastpar, Michele Ceriotti, Ola Svensson, Olivier Lévêque, Paolo Ienne, Rachid Guerraoui, Robert West, Sanidhya Kashyap, Valerio Piazza, Viesturs Simanis, Viktor Kuncak, Volkan Cevher, Philippe Schwaller, Sacha Friedli, Patrick Jermann, Tanja Käser, and Antoine Bosselut. 2024. Could ChatGPT get an Engineering Degree? Evaluating Higher Education Vulnerability to AI Assistants. *Proceedings of the National Academy of Sciences*, 121(49):e2414955121. ArXiv:2408.11841 [cs].

Kawin Ethayarajh, Yejin Choi, and Swabha Swayamdipta. 2022. Understanding dataset difficulty with $\mathcal{V}$-usable information. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 5988–6008. PMLR.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models.

Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*.

Qwen Team. 2025. Qwen3 technical report.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2024. Direct preference optimization: Your language model is secretly a reward model.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Maarten Bosma, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain of thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*.

A DPO Alignment Evaluation (Details from the Lighteval Output)

On the held-out set of $N = 50$ prompt–response pairs, the policy selects the preferred answer in 38 cases, giving a reward–accuracy of 0.76 ± 0.06. The mean reward–gap between *chosen* and *rejected* completions is $+1.25$ (standard deviation 2.37); five items show a negative gap, with the worst at -5.56. Reward–gap size and accuracy are positively correlated ($\rho = 0.69$), indicating that the learned reward signal is directionally useful but imperfectly calibrated. Across the 34 distinct function names in the sample, performance is bimodal: 24 functions are solved on every occurrence, whereas 10 are never solved. Failures involve either dual-iterator logic or recursive numeric sequences, patterns that are sparsely represented in the preference data.

B DPO Augmentation Experiments

To address the limited size of our core preference dataset (n=398), we investigated two augmentation strategies using larger, publicly available datasets. The objective was to determine if incorporating broader data could improve the model's generalization capabilities without compromising its specialization.

First, we experimented with the **UltraFeedback** dataset, combining our 398 pairs with a

mathematics-focused subset of 60,917 pairs, totaling 61,315 entries. Although this dataset was thematically relevant, its inclusion resulted in a notable decrease in reward model accuracy, as shown in Figure 6. This suggests that even within the same domain, differences in prompt style and response distribution can introduce noise that harms performance on a specialized evaluation set.

Second, we evaluated the **Stanford Human Preferences (SHP)** dataset, filtered for the 'askengineers', 'askphysics', and 'askscience' subreddits. The hypothesis was that these forums would provide examples of naturalistic, human-to-human problem-solving discourse. The resulting dataset was substantial, with 107,556 entries. However, qualitative analysis revealed a significant domain mismatch; many entries consisted of general inquiries or career discussions rather than the targeted academic problem-solving format. The extensive training time combined with poor preliminary results confirmed that this dataset was unsuitable.

Ultimately, both augmentation experiments demonstrated that expanding the training data with less-aligned, albeit larger, datasets degraded performance on our specific task. This validated our final decision to train the model exclusively on the small, highly curated preference set to maintain maximal domain relevance.

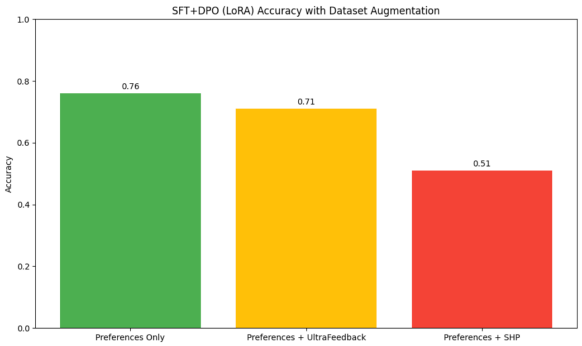

Figure 6: Effect of dataset augmentation on reward model accuracy for the SFT+DPO (LoRA) model. Augmenting the curated preference set with broader datasets like UltraFeedback and SHP led to a significant degradation in performance.

C DPO Experimental Details

The SFT and DPO training stages were configured with the following hyperparameters. These values were selected based on common practices for LoRA-based fine-tuning of language models in this parameter range and internal testing.

Table 2: Hyperparameters for SFT and DPO Training.

Parameter	Value
SFT Stage	
Learning Rate	2×10^{-4}
Batch Size	2
Gradient Accumulation	4
Number of Epochs	1
DPO Stage	
Learning Rate	5×10^{-5}
Beta (β)	0.1
Batch Size	1
Gradient Accumulation	8
Number of Epochs	1
LoRA Configuration (Both Stages)	
Rank (r)	16
Alpha (α)	32
Target Modules	q_proj, k_proj, v_proj, o_proj gate_proj, up_proj, down_proj

The DPO β parameter of 0.1 was chosen as it offers a standard balance between fitting the preference data and not diverging too far from the initial SFT model's learned distribution, which helps prevent mode collapse.

C.1 Sanfoundry MCQA Dataset Overview

The following table enumerates all multiple-choice question datasets collected from Sanfoundry, covering a wide range of STEM topics. For our analysis, we define major subjects as foundational courses in mathematics, physics, chemistry, computer science, engineering, and life sciences (see below for full list). All remaining topics are considered minor subjects.

Table 3 enumerates all MCQA datasets collected from Sanfoundry. Major subjects (**bold**) are those used in our main experiments, based on coverage of fundamental STEM disciplines (mathematics, physics, chemistry, CS, engineering, life science). Minor subjects (italic) were used for supplementary analysis.

The following is the complete list of subject datasets used for training the MCQA model in our experiments.

C.2 Dataset Classification

Major subjects are selected for their broad representation of core STEM topics. Minor subjects are used for supplemental evaluation. The full set contains over 100 MCQA datasets spanning engineering, science, and computing.

Table 3: Sanfoundry MCQA Datasets by Subject Area

Subject Area	Dataset Filenames
Mathematics	mathematics_mcqs_-_class_10 mathematics_mcqs_-_class_11 mathematics_mcqs_-_class_12 engineering_mathematics discrete_mathematics probability_and_statistics numerical_methods mathematics_of_data_mcqa
Physics	physics_mcqs_-_class_11 physics_mcqs_-_class_12 engineering_physics_i engineering_physics_ii thermodynamics heat_transfer fluid_mechanics strength_of_materials mechanical_behaviour mechanical_metallurgy engineering_mechanics structural_analysis
Chemistry	chemistry_mcqs_-_class_11 chemistry_mcqs_-_class_12 applied_chemistry organic_chemistry biochemistry chemical_process_calculation chemical_reaction_engineering solid_state_chemistry
Computer Science	artificial_intelligence data_science_mcqa cryptograph_&_network_security data_structures_&_algorithms_i data_structures_&_algorithms_ii computer_architecture computer_fundamentals computer_network cloud_computing neural_networks operating_system compiler database_management software_engineering digital_circuits microprocessor python object_oriented_programming linux unix
Engineering	automata_theory_mcqa electric_circuits electromagnetic_theory analog_circuits digital_circuits signals_and_systems control_systems power_systems electronic_devices_&_circuits network_theory electrical_machines
Life Sciences	biology_mcqs_-_class_11 biology_mcqs_-_class_12 life_sciences cell_biology

D RAG Corpus OCW Categories

Table 4: RAG Corpus OCW Categories

Category	DocCount
Algebra and Number Theory	297
Applied Mathematics	91
Calculus	12
Computer Science	1055
Differential Equations	131
Electrical Engineering	16
Electronics	66
Linear Algebra	95
Materials Science and Engineering	20
Mathematical Analysis	6
Mathematics	245
Physics	360
Probability and Statistics	165
Science	689
Systems Engineering	145

D.1 Initial Compared Evaluation

Table 5: Model Accuracy Comparison for MCQA and RAG Variants

Initial Compared Evaluation			
Model / Args	k	Dataset	Accuracy
Qwen-3 (0.6B-base)	5	Demo50	0.38
Qwen-3 (0.6B-base)	5	NLP4Edu	0.39
trip1ech/NLP4Education_mcqs	no RAG	NLP4Edu	0.4806
trip1ech/MCQA-experiment	5	NLP4Edu	0.4893
nililba/MNLP_M2_rag_model	5	NLP4Edu	0.4653
nililba/MNLP_M2_rag_model	5	NLP4Edu	0.4572
trip1ech/NLP4Education_mcqs	no RAG	Demo50	0.44
trip1ech/MCQA-experiment	7	Demo50	0.42
trip1ech/MCQA-experiment	5	Demo50	0.44
RAG Trainings Compared Evaluation			
Model Args / Name	k	Dataset	Accuracy
Qwen/Qwen3-0.6B (RAG)	5	NLP4Edu	0.39
nililba/MNLP_M2_rag_model	5	NLP4Edu	0.4653
nililba/MNLP_M2_rag_model	7	NLP4Edu	0.4572
nililba/MNLP_M2_rag_model	3	NLP4Edu	0.4638
nililba/rag-sampled	5	NLP4Edu	0.25
nililba/rag-2point	5	NLP4Edu	0.4195
Qwen/Qwen3-0.6B (RAG)	5	Demo50	0.38
nililba/MNLP_M2_rag_model	5	Demo50	0.36
nililba/MNLP_M2_rag_model	7	Demo50	0.42
nililba/MNLP_M2_rag_model	5	Demo50	0.44
nililba/rag-sampled	5	Demo50	0.26
trip1ech/MCQA-experiment	5	Demo50	0.44
trip1ech/MCQA-experiment	7	NLP4Edu	0.46
trip1ech/MCQA-experiment	4	NLP4Edu	0.4934
trip1ech/MCQA-experiment	5	NLP4Edu	0.4893
trip1ech/MCQA-experiment	6	NLP4Edu	0.4908
trip1ech/MCQA-experiment	7	NLP4Edu	0.4929
trip1ech/MCQA-experiment	8	NLP4Edu	0.4883
trip1ech/MCQA-experiment	9	NLP4Edu	0.4898
Final RAG Model Evaluation			
Model Args / Name	Dataset	k	Accuracy
trip1ech/MCQA-experiment	NLP4Edu	no RAG	0.4893
nililba/MNLP_M3_rag_model	NLP4Edu	5	0.502
nililba/MNLP_M3_rag_model	NLP4Edu	7	0.4954
nililba/MNLP_M3_rag_model	NLP4Edu	1	0.4929
nililba/MNLP_M3_rag_model	NLP4Edu	4	0.4929
nililba/MNLP_M3_rag_model	NLP4Edu	6	0.4924
nililba/MNLP_M3_rag_model	Demo50	5	0.44
nililba/MNLP_M3_rag_model	Demo50	7	0.46
nililba/MNLP_M3_rag_model	Demo50	4	0.46
nililba/MNLP_M3_rag_model	Demo50	3	0.48